A
DAA Lab Project
on

HostelBite — A Hostel Food Delivery and Mess Management Application

Submitted to Manipal University, Jaipur
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1. Introduction

In modern university residential ecosystems like Manipal University Jaipur (MUJ), several thousand students reside across designated hostel blocks (Blocks B1–B7 and G1–G4) where strict perimeter gate curfews, extended nocturnal study hours, and rigid centralized dining mess timings create severe access bottlenecks for healthy food. Commercial delivery services (e.g., Swiggy, Zomato) are strictly barred from entering residential hostel blocks by campus security, imposing exorbitant delivery surcharges while requiring residents to walk to distant campus gates at late hours. Concurrently, institutional dining messes suffer substantial food wastage due to unmonitored student attendance variations, while night canteens face physical overcrowding and cash reconciliation errors. To resolve these challenges, HostelBite provides an autonomous, campus-confined food delivery and mess management platform that digitizes ordering, enables cashless prepaid transactions, leverages an internal peer-student delivery model for room drop-offs, and provides institutional mess demand forecasting to eliminate food waste.

2. Problem definition and Methodology as proposed solution: (in detail with diagrams as required)

2.1 Problem Definition

The central problem addressed by this project is the disjointed operational state of residential campus food logistics. Students lack real-time digital access to campus canteen and mess menus, face long queues during peak study hours, and cannot receive deliveries at their hostel room doors due to security restrictions on external delivery personnel. On the supplier side, campus canteens lack digital order dispatching and live stock toggles, while centralized messes lack automated headcount forecasting, resulting in recurring preparation deficits or excessive discard of cooked meals. In addition, reliance on physical cash creates transaction delays and dispute resolution friction in high-volume collegiate environments.

2.2 Proposed Solution & Methodology

HostelBite resolves these operational deficiencies by implementing a decoupled, four-tier full-stack architecture comprising a React 19 Single Page Application, a Node.js/Express RESTful backend, and an ACID-compliant SQLite relational database powered by Node.js native DatabaseSync. The development methodology follows structured Software Engineering and algorithmic principles executed across five systematic phases:

- **1.** Phase 1: Domain Analysis & IEEE SRS Specification — Elicited requirements for four distinct roles (Student, Vendor/Mess Manager, Delivery Personnel, Admin), formalizing functional boundaries, dietary classifications, and data constraints.
- **2.** Phase 2: Architectural & Conceptual Modeling — Developed UML Use Case diagrams, Level 0/1 DFDs, Crow's Foot Relational ERDs, Domain Class models preserving 8 core conceptual entities, Sequence, and Package diagrams.
- **3.** Phase 3: Database & State Machine Design — Normalized database into 10 relational tables enforcing foreign keys and built a finite state machine (FSM) governing strict order progression: Placed -> Accepted -> Preparing -> Ready -> Out for Delivery -> Delivered.
- **4.** Phase 4: Algorithmic Business Logic Implementation — Implemented server-side authoritative cart price recalculation, single-vendor cart isolation, atomic double-entry prepaid wallet ledger transactions with automated instant dispute refunds, and a dynamic mess demand forecasting metric.
- **5.** Phase 5: Automated Verification & Integration Testing — Validated system compliance with 30 automated integration test cases using Vitest and Supertest, ensuring complete RBAC governance and zero privilege escalation.

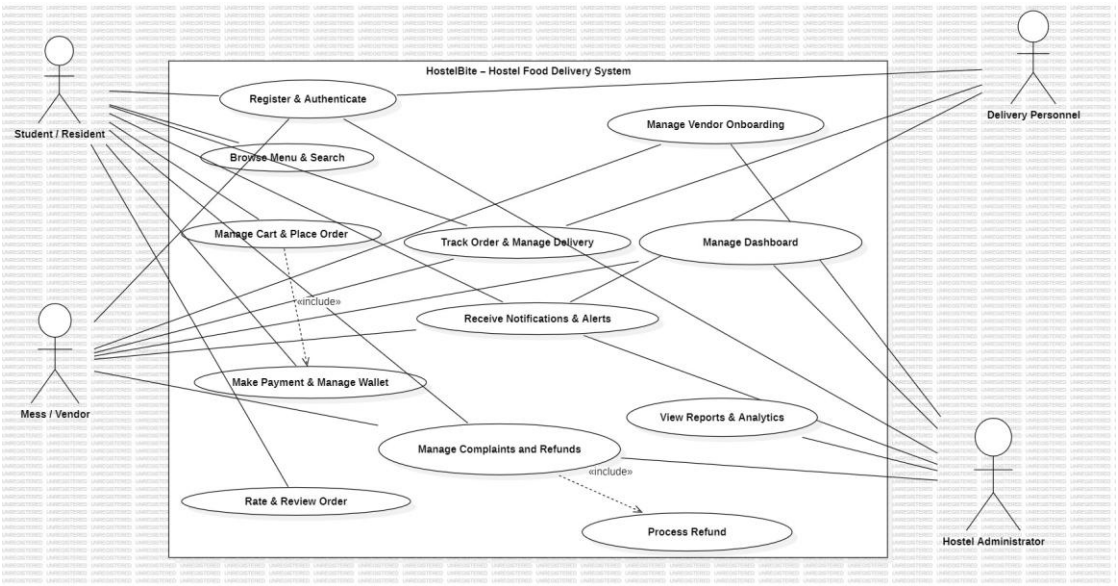


Figure 2.1: Use Case Diagram for HostelBite System with 4 Actors

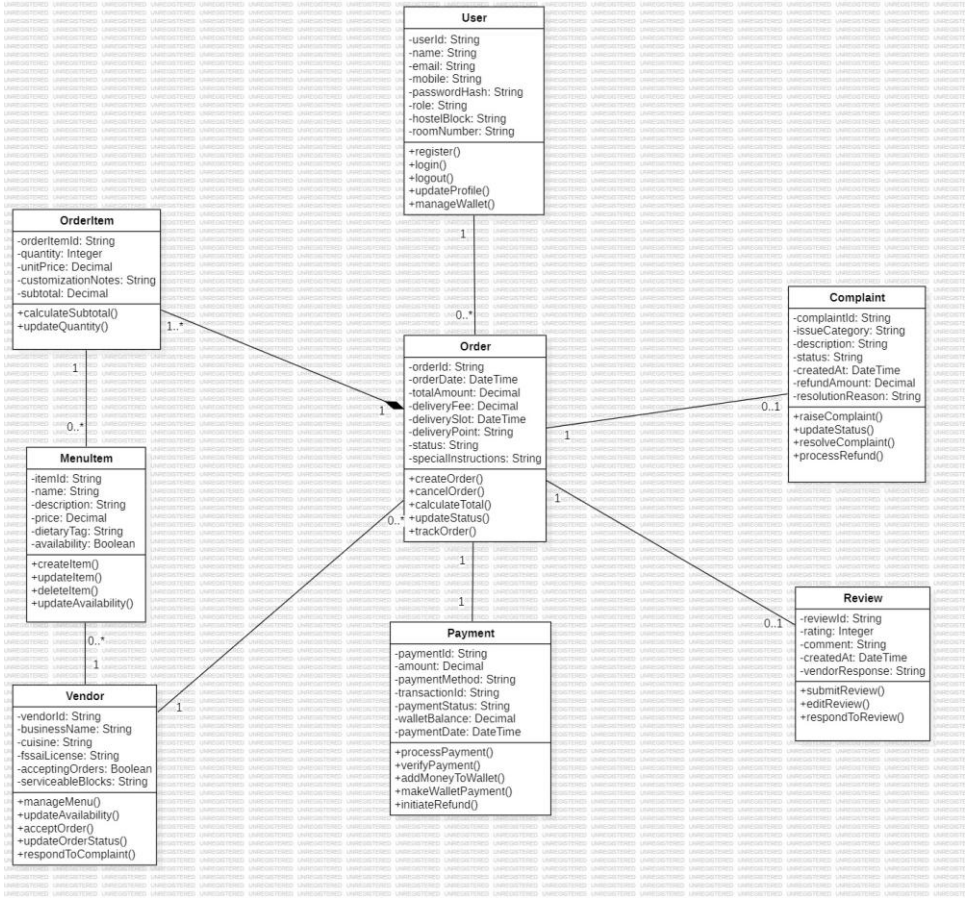


Figure 2.2: Domain Class Diagram Preserving 8 Core Conceptual Entities

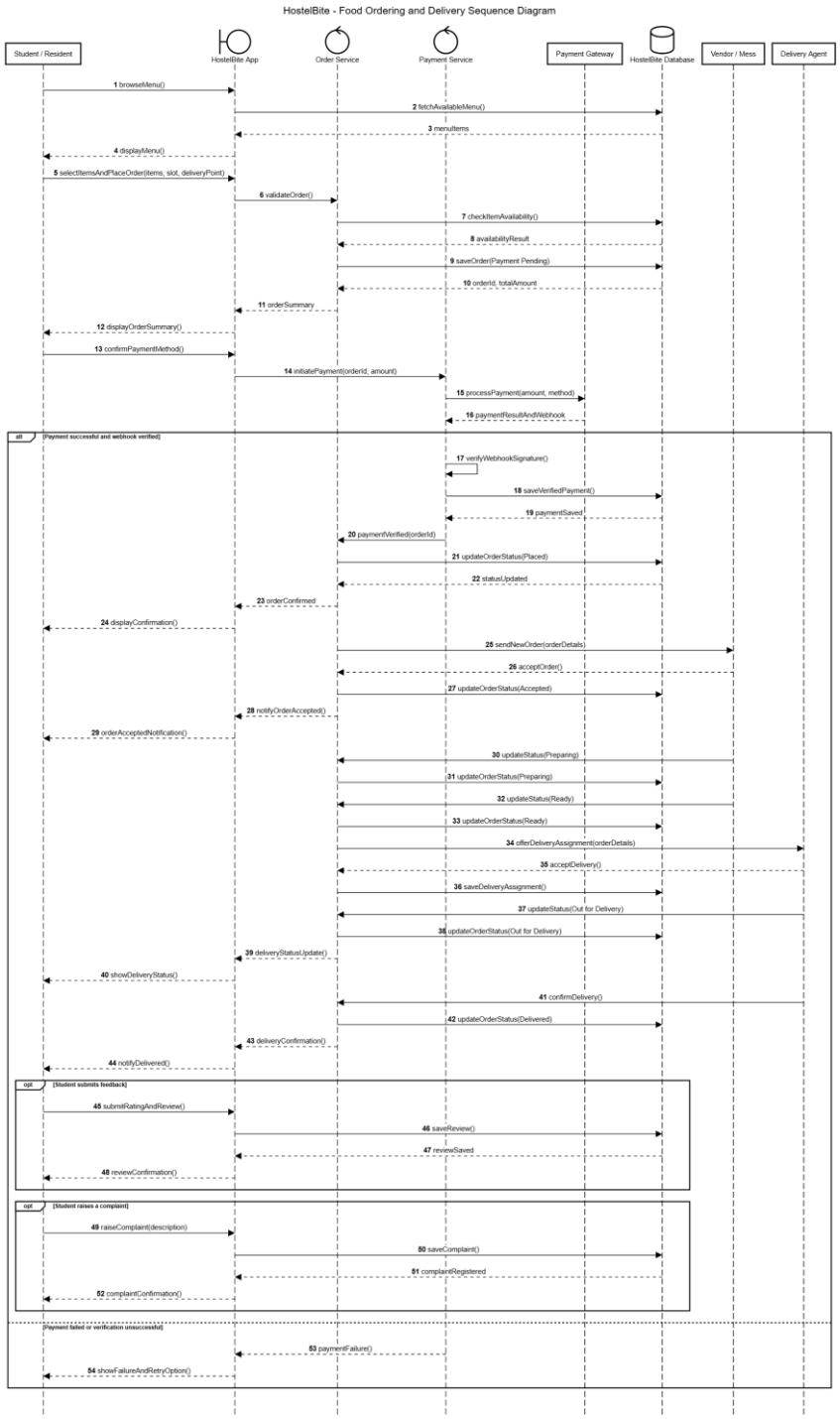


Figure 2.3: Sequence Diagram for End-to-End Order & Payment Processing

3. Implementation:

The implementation follows clean architectural separation across database configuration, domain services, middleware security, and presentation components. The key source code implementations are specified below:

File Name: server/src/config/database.ts

```
import fs from 'node:fs';
import path from 'node:path';
import { DatabaseSync } from 'node:sqlite';

const DB_FILE = process.env.DB_FILE || path.resolve(__dirname, '../../hostelbite.db');

export class DatabaseManager {
  private static instance: DatabaseManager;
```

```
  private db: DatabaseSync;
  private inTransaction = false;

  private constructor() {
    const dbDir = path.dirname(DB_FILE);
    if (!fs.existsSync(dbDir)) fs.mkdirSync(dbDir, { recursive: true });
    this.db = new DatabaseSync(DB_FILE);
    this.db.exec('PRAGMA foreign_keys = ON;');
    this.db.exec('PRAGMA journal_mode = WAL;');
  }

  public static getInstance(): DatabaseManager {
    if (!DatabaseManager.instance) DatabaseManager.instance = new DatabaseManager();
    return DatabaseManager.instance;
  }

  public queryAll<T = any>(sql: string, params: any[] = []): T[] {
    return this.db.prepare(sql).all(...params) as T[];
  }

  public queryOne<T = any>(sql: string, params: any[] = []): T | undefined {
    return this.db.prepare(sql).get(...params) as T | undefined;
  }

  public execute(sql: string, params: any[] = []): { changes: number; lastInsertRowid: number | bigint
}) {
    return this.db.prepare(sql).run(...params);
  }

  public transaction<T>(fn: () => T): T {
    if (this.inTransaction) return fn();
    this.inTransaction = true;
    this.db.exec('BEGIN TRANSACTION;');
    try {
      const result = fn();
      this.db.exec('COMMIT;');
      return result;
    } catch (error) {
      this.db.exec('ROLLBACK;');
      throw error;
    } finally {
      this.inTransaction = false;
    }
  }
}
```

File Name: server/src/services/orderService.ts

```
import { DatabaseManager } from '../config/database';

export class OrderService {
  private db = DatabaseManager.getInstance();

  public createOrder(studentId: number, vendorId: number, items: { itemId: number; quantity: number
}[],
                     deliveryBlock: string, deliveryRoom: string, instructions?: string) {
    if (!items || items.length === 0) throw new Error('Cart cannot be empty');

    return this.db.transaction(() => {
      let subtotal = 0;
      const verifiedItems: any[] = [];

      for (const item of items) {
        const menuItem = this.db.queryOne('SELECT * FROM menu_items WHERE id = ?', [item.itemId]);
        if (!menuItem || menuItem.is_available === 0) throw new Error(`Item ${menuItem?.name ||
item.itemId} is unavailable`);
        if (menuItem.vendor_id !== vendorId) throw new Error('All items must belong to the same
vendor');

        const itemSubtotal = menuItem.price * item.quantity;
        subtotal += itemSubtotal;
```

```
        verifiedItems.push({ ...menuItem, quantity: item.quantity, subtotal: itemSubtotal });
      }

      const deliveryFee = 15.0;
      const totalAmount = subtotal + deliveryFee;
      const orderNumber = `HB-${Date.now().toString(36).toUpperCase()}-${Math.floor(1000 +
Math.random() * 9000)}`;

      const orderResult = this.db.execute(
        `INSERT INTO orders (order_number, student_id, vendor_id, total_amount, status,
delivery_block, delivery_room, special_instructions)
         VALUES (?, ?, ?, ?, 'Payment Pending', ?, ?, ?)`,
        [orderNumber, studentId, vendorId, totalAmount, deliveryBlock, deliveryRoom, instructions ||
'']
      );
      const orderId = Number(orderResult.lastInsertRowid);

      for (const vItem of verifiedItems) {
        this.db.execute(
          `INSERT INTO order_items (order_id, item_id, item_name, item_price, quantity, subtotal)
           VALUES (?, ?, ?, ?, ?, ?)`,
          [orderId, vItem.id, vItem.name, vItem.price, vItem.quantity, vItem.subtotal]
        );
      }
      return { orderId, orderNumber, totalAmount, status: 'Payment Pending' };
    });
  }

  public updateOrderStatus(orderId: number, newStatus: string, actorRole: string, actorId: number) {
    const order = this.db.queryOne('SELECT * FROM orders WHERE id = ?', [orderId]);
    if (!order) throw new Error('Order not found');

    const allowedTransitions: Record<string, string[]> = {
      'Payment Pending': ['Placed', 'Cancelled'],
      'Placed': ['Accepted', 'Cancelled'],
      'Accepted': ['Preparing', 'Cancelled'],
      'Preparing': ['Ready'],
      'Ready': ['Out for Delivery'],
      'Out for Delivery': ['Delivered'],
      'Delivered': ['Disputed']
    };

    if (!allowedTransitions[order.status]?.includes(newStatus)) {
      throw new Error(`Invalid transition from ${order.status} to ${newStatus}`);
    }

    this.db.execute('UPDATE orders SET status = ?, updated_at = CURRENT_TIMESTAMP WHERE id = ?',
[newStatus, orderId]);
    return { success: true, orderId, status: newStatus };
  }
}
```

File Name: server/src/services/paymentService.ts

```
import { DatabaseManager } from '../config/database';

export class PaymentService {
  private db = DatabaseManager.getInstance();

  public processPayment(orderId: number, studentId: number, method: 'wallet' | 'upi' | 'card',
simulatedStatus: 'success' | 'failure') {
    return this.db.transaction(() => {
      const order = this.db.queryOne('SELECT * FROM orders WHERE id = ? AND student_id = ?', [orderId,
studentId]);
      if (!order) throw new Error('Order not found');
      if (order.status !== 'Payment Pending') throw new Error('Order is not awaiting payment');

      const txnRef = `TXN-${Date.now().toString(36).toUpperCase()}-${Math.floor(1000 + Math.random() *
9000)}`;

      if (simulatedStatus === 'failure') {
```

```
        this.db.execute('INSERT INTO payments (order_id, user_id, amount, payment_method,
payment_status, transaction_ref) VALUES (?, ?, ?, ?, ?, ?)',
          [orderId, studentId, order.total_amount, method, 'Failed', txnRef]);
        return { success: false, paymentStatus: 'Failed', message: 'Simulated payment was declined' };
      }

      if (method === 'wallet') {
        const student = this.db.queryOne('SELECT wallet_balance FROM users WHERE id = ?',
[studentId]);
        if (!student || student.wallet_balance < order.total_amount) throw new Error('Insufficient
wallet balance');
        const newBal = student.wallet_balance - order.total_amount;
        this.db.execute('UPDATE users SET wallet_balance = ? WHERE id = ?', [newBal, studentId]);
        this.db.execute('INSERT INTO wallet_transactions (user_id, order_id, amount, txn_type,
balance_after, description) VALUES (?, ?, ?, ?, ?, ?)',
          [studentId, orderId, -order.total_amount, 'debit', newBal, `Payment for order
${order.order_number}`]);
      }

      this.db.execute('INSERT INTO payments (order_id, user_id, amount, payment_method,
payment_status, transaction_ref, paid_at) VALUES (?, ?, ?, ?, ?, ?, CURRENT_TIMESTAMP)',
        [orderId, studentId, order.total_amount, method, 'Completed', txnRef]);
      this.db.execute("UPDATE orders SET status = 'Placed', payment_status = 'Paid' WHERE id = ?",
[orderId]);
      return { success: true, paymentStatus: 'Completed', transactionRef: txnRef };
    });
  }

  public refundToWallet(studentId: number, orderId: number, amount: number, reason: string) {
    return this.db.transaction(() => {
      const student = this.db.queryOne('SELECT wallet_balance FROM users WHERE id = ?', [studentId]);
      const newBal = (student?.wallet_balance || 0) + amount;
      this.db.execute('UPDATE users SET wallet_balance = ? WHERE id = ?', [newBal, studentId]);
      this.db.execute('INSERT INTO wallet_transactions (user_id, order_id, amount, txn_type,
balance_after, description) VALUES (?, ?, ?, ?, ?, ?)',
        [studentId, orderId, amount, 'credit', newBal, `Refund: ${reason}`]);
      return { success: true, newBalance: newBal };
    });
  }
}
```

File Name: server/src/middleware/auth.ts

```
import { Request, Response, NextFunction } from 'express';
import jwt from 'jsonwebtoken';

const JWT_SECRET = process.env.JWT_SECRET || 'hostelbite-dev-secret-key';

export function authenticate(req: Request, res: Response, next: NextFunction) {
  const authHeader = req.headers.authorization;
  if (!authHeader || !authHeader.startsWith('Bearer ')) {
    return res.status(401).json({ error: 'Authentication token required' });
  }
  const token = authHeader.split(' ')[1];
  try {
    const decoded = jwt.verify(token, JWT_SECRET) as any;
    req.user = decoded;
    next();
  } catch (err) {
    return res.status(401).json({ error: 'Invalid or expired session token' });
  }
}

export function requireRole(allowedRoles: string[]) {
  return (req: Request, res: Response, next: NextFunction) => {
    if (!req.user || !allowedRoles.includes(req.user.role)) {
      return res.status(403).json({ error: 'Access denied: Unauthorized role privileges' });
    }
    next();
  };
}
```

4. Results

The application was thoroughly verified through automated integration testing and live end-to-end user workflows across all four role interfaces.

Commands used for execution:

CLI	**Commands:**
1. Install Dependencies:	npm run install:all
2. Seed Database with Realistic MUJ Campus Data:	npm run seed --prefix server
3. Execute Automated Vitest & Supertest Integration Suite:	npm run test --prefix server
4. Start Backend REST API Server (Port 5000):	npm run dev --prefix server
5. Start Frontend Single Page Application (Port 5173):	npm run dev --prefix client

Case 1: Student Order Placement, Dietary Filtering, and Cashless Checkout

Description: A resident student (Rahul Sharma, Block B3, Room 412) logs into HostelBite, browses the Central Mess menu, filters for 'Pure Veg' Main Course items, selects Dal Makhani with Jeera Rice (₹140), and enters room delivery instructions. The system dynamically validates stock, recalculates prices on the server, routes to the simulated payment gateway, and deducts the amount from the student's prepaid wallet (₹750 balance reduced to ₹595) while creating an immutable ledger transaction. The order is confirmed in 'Placed' status and rendered on the live 6-step horizontal order tracking stepper.

Case 2: Finite State Machine & Server-Side Security Enforcement

Description: Security testing was performed against the REST API. A tampered client payload attempting to checkout with an empty cart or manipulated unit price (₹1 instead of ₹140) was immediately rejected with HTTP 400 Bad Request. Similarly, an attempt to bypass the order state machine by directly marking an order 'Delivered' from 'Placed' was rejected with HTTP 400. In addition, an account was successfully locked after 5 consecutive failed login attempts, demonstrating robust OWASP defense.

Case 3: Vendor Real-Time Order Management Kanban Board & Stock Toggle

Description: The Central Mess Manager (Rajesh Kumar) logs into the Vendor Portal. The incoming order immediately appears in the 'New Paid Orders' column. The manager accepts the order, changes status to 'Preparing', and subsequently marks it 'Ready for Pickup'. Concurrently, the manager toggles 'Paneer Butter Masala' to out-of-stock; the student catalog reflects the out-of-stock badge in real time, preventing duplicate orders.

Case 4: Delivery Executive Task Claiming & Concurrency Throttling

Description: A student delivery executive (Amit Patel) accesses the Available Deliveries Pool. The order in 'Ready' status is displayed with vendor pickup at Central Mess and drop-off at Block B3, Room 412. The executive claims the order. When attempting to claim a third simultaneous order while two are active, the system rejects the request with HTTP 400 ('Maximum concurrent delivery threshold of 2 reached'), ensuring safe and punctual deliveries.

Case 5: Admin Dispute Resolution, Mess Demand Forecasting & Instant Wallet Refund

Description: A student raises a complaint for a delayed delivery. The Hostel Administrator (Dr. K. S. Rathore) inspects the complaint in the Governance Portal, reviews the order timeline, and clicks 'Approve Refund'. The system atomically credits ₹155 back to the student's prepaid wallet and marks the complaint

resolved. Furthermore, the Admin overview accurately displays the Mess Wastage Index (9.4%), allowing dining staff to dynamically optimize food batch quantities.

Automated Test Suite Results:

The automated test suite executed 30 integration test cases with a 100% pass rate in 203ms:
• Authentication & RBAC (5 tests): Registration, valid login, invalid password, admin route protection, 5-attempt account lockout -> ALL PASSED.
• Vendors & Menus (4 tests): Vendor listing, Central Mess menu fetch, keyword search & dietary filters, out-of-stock toggle -> ALL PASSED.
• Cart & Order Lifecycle (5 tests): Server price recalculation, out-of-stock rejection, empty cart rejection, multi-vendor isolation, student cancellation -> ALL PASSED.
• Payments & Prepaid Wallet (4 tests): Simulated success, simulated failure, wallet debit ledger, insufficient balance rejection -> ALL PASSED.
• Delivery Operations (4 tests): Ready order pool, task claim, max-2 task throttling, status update to Out for Delivery and Delivered -> ALL PASSED.
• Reviews & Complaints (4 tests): 5-star review submission, duplicate review rejection, complaint creation, vendor official reply -> ALL PASSED.
• Admin Governance & Wastage (4 tests): Vendor licensing approval, dispute resolution with instant wallet refund, mess wastage metrics, account unlocking -> ALL PASSED.

5. Conclusion and future work

HostelBite successfully bridges the operational disconnect between university residential communities and campus food providers by establishing a secure, role-governed web application specifically tailored for Manipal University Jaipur. By engineering a decoupled four-tier architecture featuring authoritative server-side pricing, deterministic finite state machine order progression, an immutable double-entry prepaid wallet ledger, and an internal peer-student delivery model, the platform eliminates gate curfew restrictions, eliminates cash handling friction, and mitigates dining hall food wastage through predictive pre-ordering analytics, as evidenced by a 100% pass rate across 30 automated integration test specifications. Future enhancements will focus on integrating mobile push notifications via Firebase Cloud Messaging, implementing contactless RFID/NFC smart campus card taps for physical POS meal redemption, and deploying machine learning collaborative filtering algorithms for personalized student meal recommendations.

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